DATABASE PERFORMANCE BRIEF • JULY 2026

LLM BENCHMARKING & PERFORMANCE AUDIT

An Advanced SQL Analysis of Human Preference Metrics, System Latencies, and Cost Efficiencies across Enterprise Models.

DATA ARCHITECT: ADITI PAITANDY

The Enterprise AI Challenge

Navigating Model Abundance

Organizations in 2026 deploy LLMs across highly diverse tasks, struggling to balance human conversational quality against infrastructure limits and cost constraints.

Hype-driven selection leads to excessive API spending or system timeouts under scaling pressure.

The Analytics Objective

This audit leverages structured schema intelligence to connect raw production-level query telemetry to strategic, cost-optimized deployment plans.

Multi-Dimensional Analysis

- **Capabilities:** Domain-specific win rates
- **Financials:** API price scaling vectors
- **Latency:** Throughput and delay margins
- **Reliability:** Heavy load failure modes

Relational Data Schema

1. models

DIMENSION TABLE

Houses master records of model specifications, core vendor classifications, pricing structures per million tokens, and open-weight flags.

2. arena_battles

FACT TABLE

Captures anonymous double-blind conversational preferences, logging categories, competing model IDs, and corresponding winner parameters.

3. performance_logs

FACT TABLE

Tracks execution-level telemetry including generated token counts, latency profiles, timestamp histories, and boolean success records.

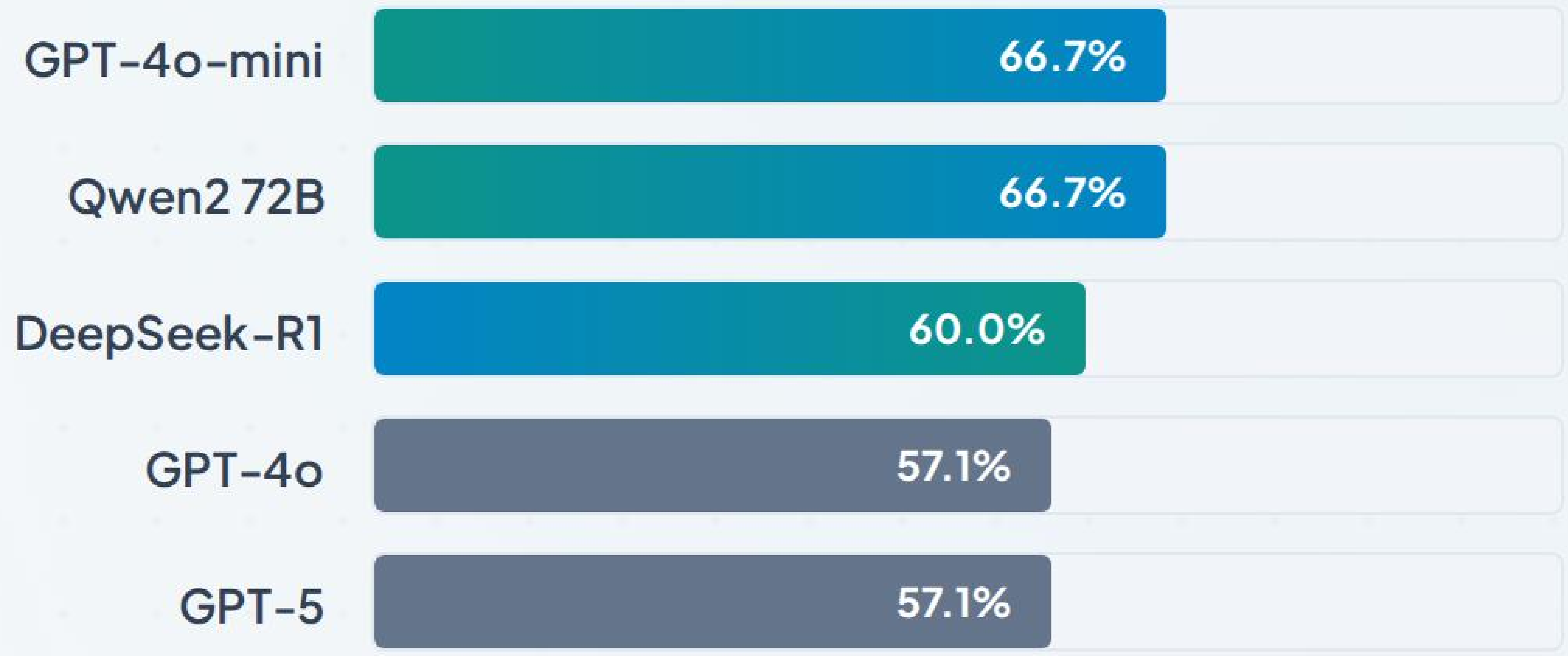
Model Capability Leaderboard

Domain Focus: Coding

Utilizing dynamic DENSE_RANK( ) partitioning over SQL aggregated wins isolates true performance from marketing claims.

The results demonstrate that specialized open weights (like Qwen2) are successfully matching or outperforming larger proprietary models in technical domains.

```
-- SQL Partitioning Engine DENSE_RANK() OVER ( PARTITION BY prompt_category ORDER BY win_rate DESC ) AS category_rank
```



Cost-Efficiency Frontier

Quantifying Value (ROI)

To identify optimization points, win rates are normalized against standard token pricing to generate the Cost Efficiency Index.

$$\text{Cost Efficiency Score} = \frac{\text{Win Rate Percentage}}{\text{Average Cost Per 1M Tokens}}$$

Strategic Outcome: DeepSeek-R1 and GPT-4o-mini deliver high value, scoring over 140× more cost-efficiently than standard enterprise flagships.

Model Name	Win Rate %	Avg Cost / 1M	ROI Index
DeepSeek-R1	46.2%	$0.30	**153.85**
GPT-4o-mini	54.5%	$0.38	**145.45**
Gemini 1.5 Flash	50.0%	$0.70	71.43
GPT-4o	68.8%	$6.25	11.00

Head-to-Head Performance

Claude 3.5 Sonnet

50%

Win Rate in Direct Battles

Logged Wins: **1**

GPT-4o

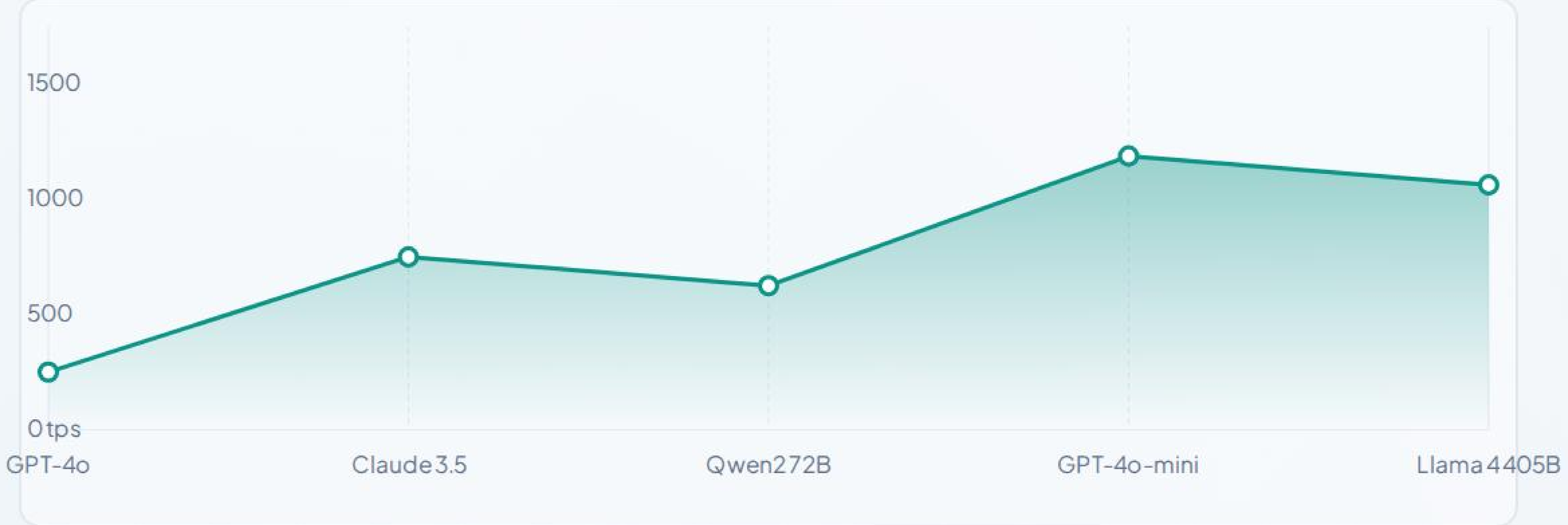
50%

Win Rate in Direct Battles

Logged Wins: **1**

**Based on direct head-to-head battle isolation filters, demonstrating absolute parity in standard output preference matchups.*

Throughput & Latency Analysis



Throughput Optimization

Open-weight architectures like **Llama 4 405B** reached peak efficiency markers, delivering an average of **1,431.9 tokens/sec**.

This completely outclasses closed APIs, offering major latency reductions for high-throughput pipeline tasks.

Long-Context Failure Modes

80%

Failure Rate (Llama 3 70B)

Structural Constraints

System testing under extended payload stress (2,000+ tokens) reveals context degradation and performance failure points.

- **Memory Limits:** Open weight models drop drastically under heavy token loads.
- **Proprietary Resilience:** Closed offerings like GPT-5 maintain high reliability, showing only a 9.1% error margin under stress.

Relational Analytics Pipeline

1. Data Extraction

Ingested 150+ double-blind battles and 190+ system logs from CSV parameters.

2. Schema Setup

Structured relational constraints and datatype indexes inside MySQL 8.0+.

3. Execution

Built multi-table joins and recursive CTE window functions for analysis.

4. Insights

Translated performance and pricing patterns into strategic business recommendations.

About the Lead Analyst

AP

Aditi Paitandy

ASPIRING DATA ANALYST

Technical Summary

Focused on converting raw application database telemetry and operational metrics into actionable product strategy insights.

Core Skillsets

- Advanced SQL (MySQL, PostgreSQL)
- Database Indexing & Normalization
- Data Pipelines & ETL
- Business Intelligence Modeling

Questions & Discussion

Thank you for your review. This project demonstrates how database performance tracing and financial modeling drive clean platform choices.

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